

Chengze “Cheng” Liu

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MS CS (Data Analytics) | Python, SQL, Machine Learning, Cloud Deployment

Seeking Data Science / ML / Cloud Internships – Spring/Summer 2027

Education

University of Houston

Houston, TX

MS in Computer Science — Data Analytics Track **GPA: 4.0**

Expected May 2027

BS in Computer Science | Minor in Mathematics **GPA: 4.0**

Dec 2025

Skills

Data & Analytics	Python, SQL, Pandas, NumPy, SciPy, Scikit-learn
Machine Learning	Regression, Classification, Clustering, Cross-Validation, Model Evaluation
Applied	NLP (TF-IDF, N-grams, POS Tagging), Deep Learning (PyTorch, TensorFlow), Computer Vision (OpenCV), Geospatial (Shapely, Ripley's L)
Cloud & Deployment	AWS (Lambda, EC2, DynamoDB, CloudWatch, S3, IAM/Security Groups), Alibaba Cloud (ECS), Azure (OAuth)
Visualization	Matplotlib, Seaborn, VTK, Scientific/3D Visualization
Web & Dev Tools	HTML, CSS, JavaScript, REST APIs, Django (basic), Git, Linux, VS Code, Jupyter Notebook, Claude Code
Languages	English (Fluent), Chinese (Fluent)

Project Experience

Robinhood Translator (AI/ML Engineering, AWS, Financial Services)

April 2026 – June 2026

- Built and deployed a Chrome extension backed by an AWS Lambda service translating live Robinhood financial content into Chinese; designed a DynamoDB caching layer with SHA-256 hashing to cut redundant API calls and reduce latency.
- Ran a model-selection pipeline benchmarking Claude Haiku, Sonnet, and a self-hosted Qwen3:8b (Ollama) on quality, latency, and cost, driving Haiku's adoption as the production model.
- Used Claude Code throughout iterative development and maintained a disciplined git commit workflow across four development iterations.

Deep Learning Framework Benchmarking (Cloud Computing, AWS)

March 2026 – April 2026

- Provisioned and benchmarked PyTorch vs. TensorFlow CNN training on AWS EC2 GPU instances (g4dn.xlarge, NVIDIA T4) across 12 runs, instrumenting CloudWatch and a custom GPU monitoring script for hardware-level performance metrics.
- Found PyTorch reached 80% CIFAR-10 accuracy 2.6× faster and cheaper than TensorFlow, tracing the gap to `@tf.function` graph-compilation overhead offsetting TensorFlow's efficient data pipeline.

Gender Prediction from Blog Posts (NLP, Machine Learning)

March 2026 – April 2026

- Achieved 72.5% gender classification accuracy (vs. 70% baseline) on 3,226 blog posts, comparing linear SVM, logistic regression, and decision tree models with TF-IDF n-gram features.
- Ran ablation experiments across word-, character-, and POS-tag features; found stylistic signals degraded accuracy, revealing topical content as the dominant gender predictor.

Spatial Collocation Analysis of Urban Building Types (Data Mining, Spatial Statistics)

March 2026 – April 2026

- Parsed a 3,371-building geospatial KML dataset and implemented Ripley's L-function and bivariate cross-L via `scipy cKDTree`, using a label-swap Monte Carlo null model (199 simulations) to reveal anti-collocation patterns that raw nearest-neighbor distances had missed.

Weather Data Outlier Detection (Data Mining, Unsupervised Learning)

Jan 2026 – Feb 2026

- Compared three unsupervised outlier detection methods — GMM, KDE, and a TensorFlow autoencoder — across 9 hyperparameter configurations on 365-day Houston weather data, validated against a synthetic dataset with known outlier structure.
- Found GMM uniquely captured rare multivariate anomalies while KDE and the autoencoder each reduced to simpler univariate patterns, informing a concrete model-selection recommendation.

Web Application (Full-Stack, SQL)

March 2025 – May 2025

- Built a full-stack web application with a SQL relational database backend, Azure OAuth authentication, and automated PDF report generation.

Work Experience

Server / Front-of-House Staff

Sept 2021 – May 2022

Haidilao Hotpot | Katy, TX — Managed order accuracy and customer flow in a high-volume, fast-paced restaurant environment.